BIG DATA ANALYTICS

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A4CS19	PCC	3	-	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. To introduce the terminology, technology and its applications
2. To introduce the concept of Analytics and Visualization
3. To demonstrate the usage of various Big Data tools and Data Visualization tools

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. **Compare** various file systems and **use** an appropriate file system for storing different types of data.
2. **Demonstrate** the concepts of Hadoop ecosystem for storing and processing of unstructured data.
3. **Apply** the knowledge of programming to process the stored data using Hadoop tools and generate reports.
4. **Connect** to web data sources for data gathering, **Integrate** data sources with hadoop components to process streaming data.
5. **Tabulate** and **examine** the results generated using hadoop components

UNIT-I

INTRODUCATION TO BIG DATA: Data and its importance, Big Data - definition, implications of Big Data, addressing Big Data implications using Hadoop, Hadoop Ecosystem

HADOOP ARCHITECTURE:

Hadoop Storage : HDFS, Hadoop
Processing : Map Reduce Framework
Hadoop Server Roles : Name Node, Secondary Name Node and Data Node, Job Tracker, TaskTracker

HDFS-HADOOP DISTRIBUTED FILE SYSTEM: Design of HDFS, HDFS Concepts, HDFS Daemons, HDFS High Availability, Block Abstraction, FUSE: File System in User Space. HDFS Command Line Interface (CLI), Concept of File Reading and Writing in HDFS.

UNIT-II

MAPREDUCE PROGRAMMING MODEL: Introduction to Map Reduce Programming model to process Big Data, key features of Map Reduce, Map Reduce Job skeleton, Introduction to Map Reduce API, Hadoop Data Types, Develop Map Reduce Job using Eclipse, bulit a Map Reduce Job export it as a java archive(.jar file).

MAPREDUCE JOB LIFE CYCLE: Understanding Mapper, Combiner, Partitioner, Shuffle & Sort and Reduce phases of Map Reduce Application, Developing Map Reduce Jobs based on the requirement using given datasets like weather dataset.

UNIT-III

INTRODUCTION TO PIG: Understanding pig and pig Platform, introduction to Pig Latin Language and Execution engine, running pig in different modes, Pig Grunt Shell and its usage.

PIG LATIN LANGUAGE –SEMANTICS –DATA TYPES IN PIG: Pig Latin Basics, Key words, Pig Data types, Understanding Pig relation, bag, tuple and writing pig relations or statements using Grunt Shell, expressions, Data processing operators, using Built in functions.

WRITING PIG SCRIPTS USING PIG LATIN: Writing pig scripts and saving them text editor, running pig scripts from command line.

UNIT-IV		
INTRODUCATION TO HIVE: Understanding Hive Shell, Running Hive, Understanding Schema on read and Schema on write. **HIVE QL DATA TYPES, SEMANTICS:** Introduction to Hive QL (Query Language), Language semantics, Hive Data Types. **HIVE DDL, DML AND HIVE SCRIPTS:** Hive Statements, Understanding and working with Hive Data Definition Languages and Manipulation Language statements, Creating Hive Scripts and running them from hive terminal and command line.		
UNIT-V		
SQOOP: Introduction to Sqoop tool, commands to connect databases and list databases and tables, command to import data from RDBMS into HDFS, Command to export data from HDFS into required tables of RDBMS. **FLUME:** Introduction to Flume agent, understanding Flume components Source, Channel and Sink. **OOZIE:** Introduction to Oozie, Understanding work flow Management.		

TEXT BOOKS:

1. Hadoop: The Definitive Guide, 4th Edition - O'Reilly Media
2. Chris Eaton, Dirk deroos et al. , “Understanding Big data ”, McGraw Hill, 2012.
3. Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley 2015.

REFERENCE BOOKS:

1. Michael Berthold, David J. Hand, "Intelligent Data Analysis”, Springer, 2007.
2. Paul Zikopoulos ,Dirk DeRoos , Krishnan Parasuraman , Thomas Deutsch , James Giles , David Corigan , "Harness the Power of Big Data The IBM Big Data Platform ", Tata McGraw Hill Publications, 2012.